FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS

PART TEST –III
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	C	A
2.	B	C	A
3.	A	B	C
4.	D	D	A
5.	B	B	C
6.	A	C	B
7.	A	C	C
8.	A	A	C
9.	C	C	A
10.	D	C	D
11.	A, B	C, D	A, B, C, D
12.	A, D	B, C	A, B, C
13.	A, C, D	A, D	A, C, D
14.	B, D	A, B, D	B, C, D
15.	A, D	A, C	A, D
1.	0	2	1
2.	0	1	6
3.	9	3	4
4.	3	9	4
5.	5	4	3

Physics

PART – I

2. Potential energy will be finite and maximum at x = 0.

5. $B_1 - B_2 = 20\ \mu T$, $B_1 + B_2 = 50\ \mu T$ and $B \propto i$

6. Use $\tau = \vec{M} \times \vec{B}$

7. Use symmetry arguments. Net $\vec{B}$ will be perpendicular to the plane of the loop

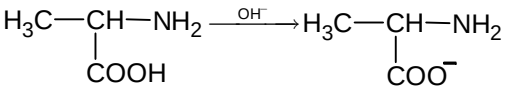
9. Use $\vec{P} = \Sigma q_i \vec{r}_i$

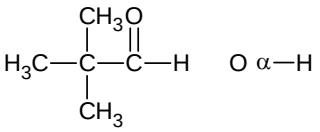
Chemistry

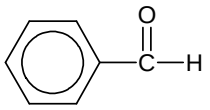
PART – II

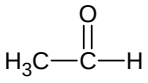
SECTION – A

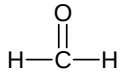
1. In alkaline medium, alanine exists as anion:

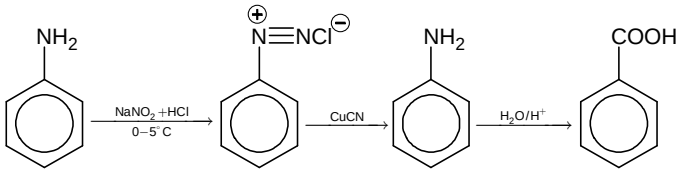


2. 
O α—H

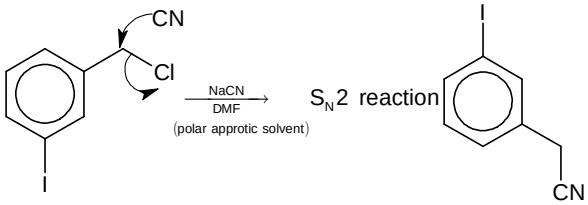

O α—H


3 α—H(not appropriate for Cannizzaro reaciton

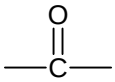

O α—H

3. 

4. Hoffmann bronamide degradation.

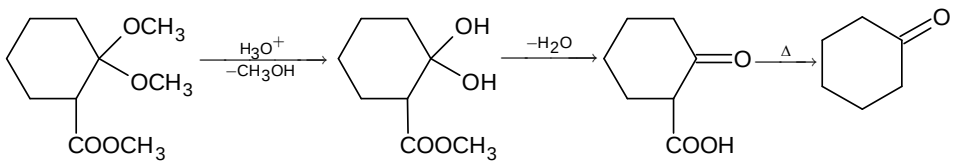
5. 

6. $NaBH_4$ selectively reduceds

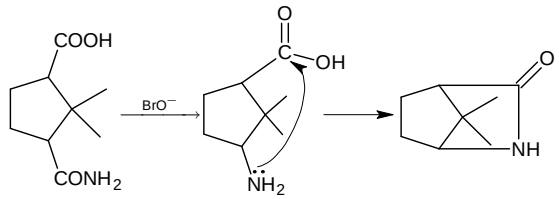


7. Gattermann Koch reaction.

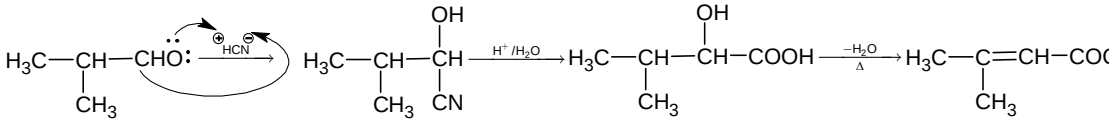
8. MnO_2 oxidized allylic alcohol.

9. 

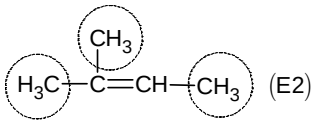
10.



11.

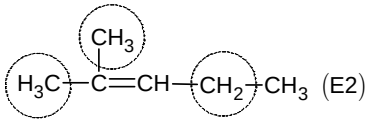


12. (A)



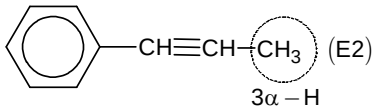
$9\alpha - H$

(B)



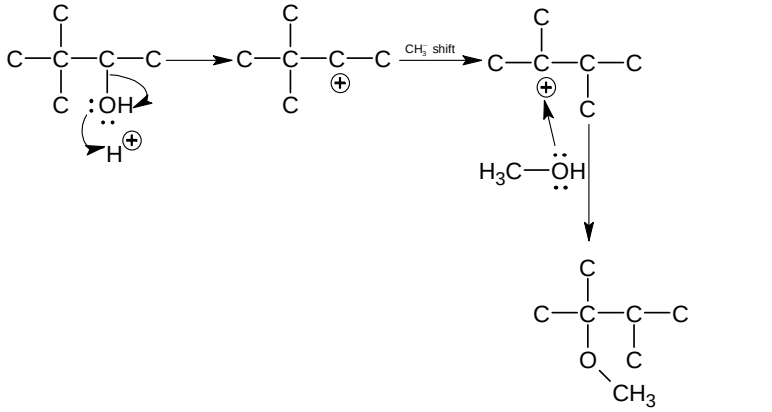
$8\alpha - H$

(C)



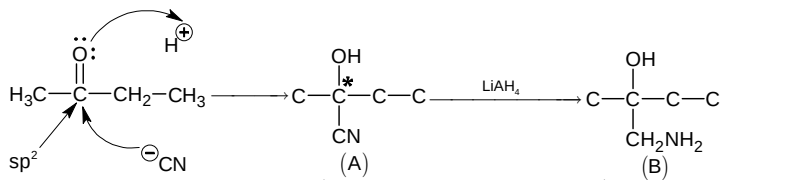
$3\alpha - H$

(D)



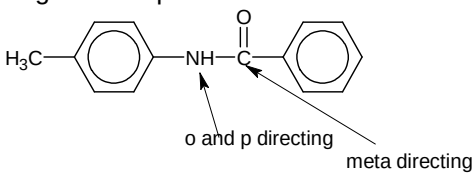
(ether and NOT an alkalene)

13.



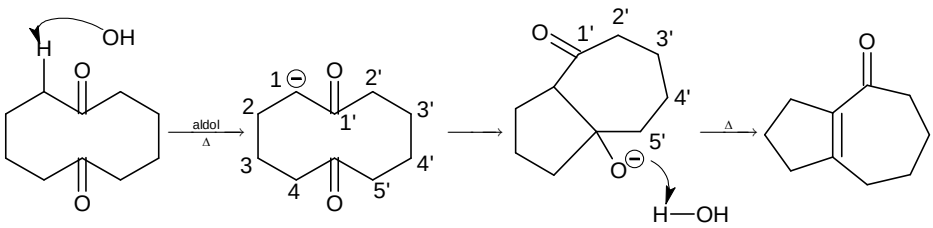
(hybridised planar) (above & below the plane) (A) (R as well as S racemic mixture) $LiAH_4$ (B) (racemic mixture)

14. Organic compound is

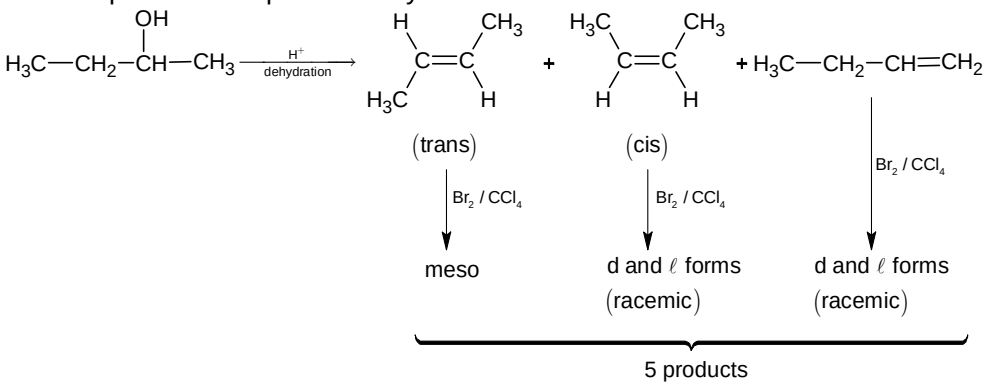


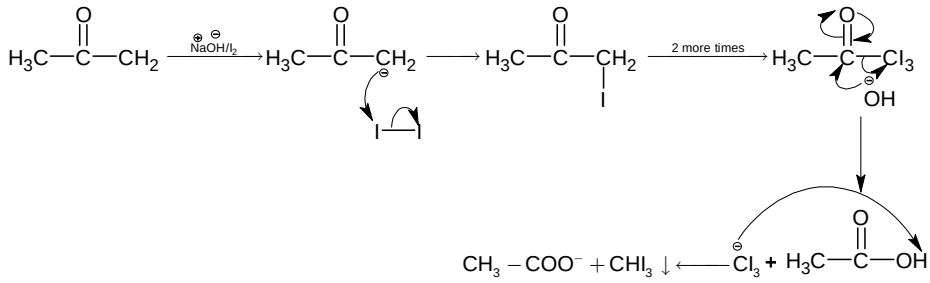
SECTION – C

1. —NH_2 si basic.
—COO^- is (conjugate) base.
—NH_3^+ is acidic.

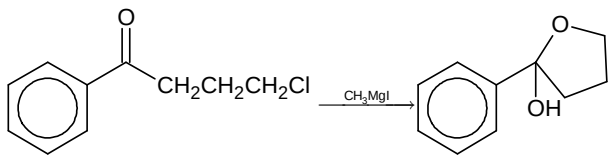
2. 

3. The compound A is represented by the 3 structures as follows:



4. 

1 molecules of $H_3C-\overset{O}{\overset{\|}{C}}-CH_3$ requires 3 molecules of I_2.

5. 

Mathematics

PART – III

SECTION – A

1. (a, 0) lies on the given curve

$\therefore\ o = \sin 2a - \sqrt{3}\sin a$

$\Rightarrow \sin a = 0$ or $\cos a = \dfrac{\sqrt{3}}{2}$

$\Rightarrow\ a = \dfrac{\pi}{6}$ (as a > 0 and 1st point of intersection with positive x-axis) and

$A = \int_0^{\pi/6} \left(\sin 2x - \sqrt{3}\sin x\right)dx\ = \dfrac{7}{4} - \sqrt{3} = \dfrac{7}{4} - 2\cos a\ \Rightarrow 4A + 8\cos a = 7$

2. Let $g(x) = (\sin x)^{\ln x} = e^{\ln x \ln(\sin x)}$

$f(x) = g'(x) = (\sin x)^{\ln x}\left[\cot x(\ln x) + \dfrac{\ln(\sin x)}{x}\right]$

Hence, $f\left(\dfrac{\pi}{2}\right) = g'\left(\dfrac{\pi}{2}\right) = 0$

3. Put $2x^{7m} + 7x^{2m} + 14x^m = t^m$

$14\ m(x^{7m-1} + x^{2m-1} + x^{m-1})\ dx = mt^{m-1}\ dt$

4. $u(x) = 7v(x) \Rightarrow p = 7$

$\Rightarrow \left(\dfrac{u(x)}{v(x)}\right)' = 0\ \Rightarrow q = 0$

$\therefore\ \dfrac{p+q}{p-q} = 1$

5. $\dfrac{dx}{dt} = -\left(\dfrac{3+2t}{t^4}\right)$

$\dfrac{dy}{dt} = -\left(\dfrac{3+2t}{t^3}\right)$

6. $f[g(x)] = x \Rightarrow f'[g(2)]\ g'(2) = 12 \Rightarrow f'(a)\ g'(2)$ [putting x = 2]

Given $f'(a) = \dfrac{a^{10}}{1+a^2}\ \Rightarrow g'(2) = \dfrac{1+a^2}{a^{10}}$

7. $f'(x) = \dfrac{1}{1+\cos x}$ integrating

$f(x) = \tan\dfrac{x}{2} + c$

$f(0) = 3\ \therefore\ c = 3$

$f(x) = \tan\dfrac{x}{2} + 3$

$\therefore\ f\left(\dfrac{\pi}{2}\right) = 4$

8. $2^y = 3^x - 1 \Rightarrow y \ln 2 = \ln(3^x - 1)$

$$\ln 2 \frac{dy}{dx} = \frac{3^x \ln 3}{3^x - 1} = \frac{\ln 3}{1 - 3^{-x}}$$

$$\lim_{x\to\infty} \frac{dy}{dx} = \frac{\ln 3}{\ln 2} \quad \lim_{x\to\infty} \frac{1}{1-3^{-x}} = \log_2 3$$

9. $I = \int (\sin 100x + x)(\sin x)^{99}\, dx = \int \left[(\sin(100x)\cos x + \cos 100x \sin x)\right](\sin x)^{99}\, dx$

$$= \int \underbrace{\sin(100x)}_{I} \underbrace{\cos x (\sin x)^{99}}_{II}\, dx + \int \cos(100x)(\sin x)^{100}\, dx$$

$$= \frac{\sin(100x)(\sin x)^{100}}{100} - \frac{100}{100}\int \cos(100x)(\sin x)^{100}\, dx + \int \cos(100x)(\sin x)^{100}\, dx$$

$$= \frac{\sin(100x)(\sin x)^{100}}{100} + c$$

10. Using Lopital Rule

$$\lim_{x\to 0}\left(\frac{a\tan^{-1}\sqrt{x}/a - b\tan^{-1}\sqrt{x}/b}{x\sqrt{x}}\right)$$

$$= \lim_{x\to 0} \frac{\frac{a}{(1+x/a^2)}\cdot\frac{1}{a}\cdot\frac{1}{2\sqrt{x}} - \frac{b}{(1+x/b^2)}\cdot\frac{1}{b}\cdot\frac{1}{2\sqrt{x}}}{\frac{3}{2}\sqrt{x}}$$

$$= \lim_{x\to 0}\left[\frac{a^2}{(a^2+x)x} - \frac{b^2}{(b^2+x)x}\right]\cdot\frac{1}{3} = \frac{a^2-b^2}{3a^2b^2}$$

13. $$f(x) = \begin{cases} x^3 \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

14. $2x - x^2 = x^n$

$\Rightarrow x = 0$ and 1

$$A = \int_0^1 \left(2x - x^2 - x^n\right)dx = x^2 - \frac{x^3}{3} - \frac{x^{n+1}}{n+1}\Big]_0^1 = \frac{2}{3} - \frac{1}{n+1}$$

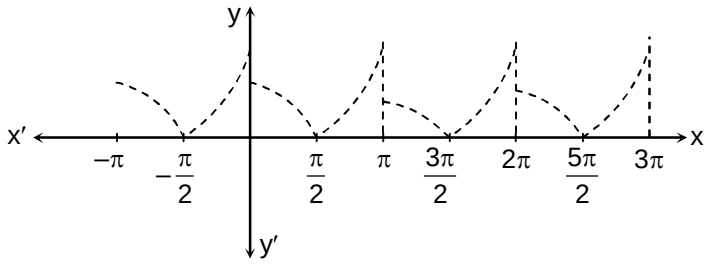
Hence, $\frac{2}{3} - \frac{1}{n+1} = \frac{1}{2}$

$\Rightarrow n = 5$

Hence, n is devisor of (20, 15, 30)

15. Given $f(x) = \begin{cases} \cos x & 0 \le x < \frac{\pi}{2} \\ \left(\frac{\pi}{2} - x\right)^2 & \frac{\pi}{2} \le x < \pi \end{cases}$ and f is periodic with period π

$\therefore$ We draw the graph of y = f(x)



From the graph the range of the function is $\left[0, \frac{\pi^2}{4}\right]$

It is discontinuous at $x = n\pi$, $n \in I$ at $x = \frac{n\pi}{2}$, $n \in I$

Area bounded by $y = f(x)$ and the x-axis from $-n\pi$ to $n\pi$ for $n \in N$

$= 2n\int_0^{\pi} f(x)dx = 2n\left[\int_0^{\pi/2} \cos x\, dx + \int_{\pi/2}^{\pi} \left(\frac{\pi}{2} - x\right)^2 dx\right] = 2n\left(1 + \frac{\pi^3}{24}\right)$

SECTION – C

1. We have $I_n + I_{n+2} = \int_0^{\pi/4} \tan^n x dx + \int_0^{\pi/4} \tan^{n+2} x dx$

$= \int_0^{\pi/4} \tan^n x\left(1 + \tan^2 x\right)dx = \int_0^1 u^n du = \frac{1}{n+1}$ (where $u = \tan x$)

So, $I_n + I_{n+2} = \frac{1}{1+n}$ which implies that $I_{n+1} + I_{n+3} = \frac{1}{n+2}$

Hence, $\sum_{n=0}^{\infty}\left(I_n I_{n+1} + I_n I_{n+3} + I_{n+1} I_{n+2} + I_{n+2} I_{n+3}\right)$

$= \sum_{n=0}^{\infty}\left(I_n + I_{n+2}\right)\left(I_{n+1} + I_{n+3}\right) = \sum_{n=0}^{\infty}\left(\frac{1}{n+1}\right)\left(\frac{1}{n+2}\right) = \sum_{n=0}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$

$\therefore\ S_n = 1 - \frac{1}{n+2} \Rightarrow \lim_{x\to\infty}\left(1 - \frac{1}{n+2}\right) = 1$

2. $L : \lim_{x\to 0} \frac{\cos^{-1}\left(\frac{\sin x}{x}\right)}{x} = \lim_{x\to 0} \frac{\sin^{-1}\left(\sqrt{\frac{x^2 - \sin^2 x}{x^2}}\right)}{x}$

$= \lim_{x\to 0} \frac{\sin^{-1}(M)}{M} \cdot \lim_{x\to 0} \frac{M}{x}$ (Using $\cos^{-1} t = \sin^{-1}\sqrt{1 - t^2}$)

$= 1 \cdot \lim_{x\to 0} \sqrt{\frac{x^2 - \sin^2 x}{x^2 \cdot x^2}} = 1 \cdot \lim_{x\to 0} \sqrt{\frac{(x - \sin x)}{x^3}\frac{(x + \sin x)}{x}}$

$= \lim_{x\to 0} \sqrt{\frac{1}{6} \cdot 2} = \frac{1}{\sqrt{3}} \Rightarrow L = \frac{1}{\sqrt{3}}$

$U : f(x) = \int_0^x \left(\sin\left(t^2 - t\right)\cos x + \cos\left(t^2 - t\right)\sin x\right)dt$

or $f(x) = \cos x \int_0^x \sin\left(t^2 - t\right)dt + \sin x \int_0^x \cos\left(t^2 - t\right)dt$

$$f'(x)=\cos x\cdot\sin\left(x^2-x\right)-\left(\int_0^x\sin\left(t^2-t\right)dt\right)\sin x+\left(\int_0^x\cos\left(t^2-t\right)dt\right)\cos x+\cos\left(x^2-x\right)\sin x$$

$$f''(x)=\left[-\sin\left(x^2-x\right)\sin x+\cos x\cdot\cos\left(x^2-x\right)(2x-1)\right]$$

$$-\left[\left(\int_0^x\sin\left(t^2-t\right)dt\right)\cos x+\sin x\sin x\left(x^2-x\right)\right]$$

$$+\left[\cos\left(x^2-x\right)\cos x-\sin x\sin\left(x^2-x\right)(2x-1)\right]$$

$$+\left[\cos x\cos\left(x^2-x\right)-\left(\int_0^x\cos\left(t^2-t\right)dt\right)\sin x\right]$$

$\therefore$ f″(0) = 1 $\therefore$ U = 1

3. $V=\frac{1}{2}\int_0^{\pi/2}x\left|2\sin^2x-1\right|dx=\frac{1}{2}\int_0^{\pi/2}x\left|\cos2x\right|dx$

Put 2x = t, $dx=\frac{dt}{2}$

4. Consider the numerator cos 4x – cos 4α
= (2 cos^2 2x – 1) – (2 cos^2 2α – 1)
= 2(cos^2 2x – cos^2 2α)
= 2(cos 2x + cos 2α)(cos 2x – cos 2α)
= 4(cos 2x + cos 2α)(cos x – cos α)(cos x + cos α)

$\therefore$ $I=4\int_0^{\pi}\left(\cos2x+\cos2\alpha\right)\left(\cos x+\cos\alpha\right)dx=4\int_0^{\pi}\left(\cos2\alpha\cdot\cos\alpha\right)dx$ (all other integral vanish)

$\therefore$ 4π cos $\alpha\cdot$cos 2α